KAIZEN MATHS

Lesson starter

Advanced Factorisation: Grouping, Difference of Squares and Cubics

HCF with Powers and Variables

Name _____

S1

$8xy - 20x^2y^3$

S2

$4x^3y - 14x^2y^3 + 2xy$

S3

$15x^3y^3 - 6x^4y^4$

S4

$12x^4y - 28x^3y^3 + 32x^2y$

Show your working

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Exit ticket

Advanced Factorisation: Grouping, Difference of Squares and Cubics

HCF with Powers and Variables

Name _____

E1

$10xy^3 - 5x^2y^4$

E2

$15x^3y - 15x^2y^3 + 5xy$

E3

$3xy^2 + 6x^3y^4$

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Teacher answers & working

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S1

$8xy - 20x^2y^3$

Answer:

$4xy(2 - 5xy^2)$

Working:

Method focus: compare coefficients and powers carefully before writing the bracket.

Step 1: Coefficients have HCF 4.

Step 2: The smallest powers are x and y.

Step 3: The full HCF is 4xy.

$8xy - 20x^2y^3 = 4xy(2 - 5xy^2)$

Step 4: Check the bracket: multiplying by the HCF gives both original terms.

S2

$4x^3y - 14x^2y^3 + 2xy$

Answer:

$2xy(2x^2 - 7xy^2 + 1)$

Working:

Method focus: the HCF must divide all three terms.

Step 1: The HCF must divide all three terms.

Step 2: Coefficients share 2, x powers share x, and y powers share y.

Step 3: The full HCF is 2xy.

$4x^3y - 14x^2y^3 + 2xy = 2xy(2x^2 - 7xy^2 + 1)$

Step 4: No further common factor remains inside the bracket.

S3

$15x^3y^3 - 6x^4y^4$

Answer:

$3x^3y^3(5 - 2xy)$

Working:

Method focus: compare coefficients and powers carefully before writing the bracket.

Step 1: Coefficients have HCF 3.

Step 2: The smallest powers are x^3 and y^3.

Step 3: The full HCF is $3x^3y^3$.

$15x^3y^3 - 6x^4y^4 = 3x^3y^3(5 - 2xy)$

Step 4: Check the bracket: multiplying by the HCF gives both original terms.

Teacher answers & working

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S4

$12x^4y - 28x^3y^3 + 32x^2y$

Answer:

$4x^2y(3x^2 - 7xy^2 + 8)$

Working:

Method focus: the HCF must divide all three terms.

Step 1: The HCF must divide all three terms.

Step 2: Coefficients share 4, x powers share x^2, and y powers share y.

Step 3: The full HCF is $4x^2y$.

$12x^4y - 28x^3y^3 + 32x^2y = 4x^2y(3x^2 - 7xy^2 + 8)$

Step 4: No further common factor remains inside the bracket.

E1

$10xy^3 - 5x^2y^4$

Answer:

$5xy^3(2 - xy)$

Working:

Method focus: compare coefficients and powers carefully before writing the bracket.

Step 1: Coefficients have HCF 5.

Step 2: The smallest powers are x and y^3.

Step 3: The full HCF is $5xy^3$.

$10xy^3 - 5x^2y^4 = 5xy^3(2 - xy)$

Step 4: Check the bracket: multiplying by the HCF gives both original terms.

E2

$15x^3y - 15x^2y^3 + 5xy$

Answer:

$5xy(3x^2 - 3xy^2 + 1)$

Working:

Method focus: the HCF must divide all three terms.

Step 1: The HCF must divide all three terms.

Step 2: Coefficients share 5, x powers share x, and y powers share y.

Step 3: The full HCF is 5xy.

$15x^3y - 15x^2y^3 + 5xy = 5xy(3x^2 - 3xy^2 + 1)$

Step 4: No further common factor remains inside the bracket.

Teacher answers & working

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E3

$3xy^2 + 6x^3y^4$

Answer:

$3xy^2(1 + 2x^2y^2)$

Working:

Method focus: compare coefficients and powers carefully before writing the bracket.

Step 1: Coefficients have HCF 3.

Step 2: The smallest powers are x and y^2.

Step 3: The full HCF is $3xy^2$.

$3xy^2 + 6x^3y^4 = 3xy^2(1 + 2x^2y^2)$

Step 4: Check the bracket: multiplying by the HCF gives both original terms.